CHAPTER 15

Application of computational approaches for the drug discovery of cholesterol gallstone disease: identification of new farnesoid X receptor modulators as a case study

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Introduction

Cholelithiasis, or gallstones, is one of the most common gastrointestinal diseases, characterized by the formation of stones in the gallbladder, bile ducts, or both (Swarne et al., 2021). Gallstones can be categorized into three types based on their appearance and composition: cholesterol gallstones, pigment gallstones (black and brown), and mixed gallstones (Uppal et al., 2008). More than 80% of gallstone patients have cholesterol gallstones, which develop due to a combination of systemic and local factors (Di Ciaula & Portincasa, 2018). The metabolic imbalance between lipids and bile salts in gallbladder bile is the primary cause of cholesterol gallstone disease (CGD) (Reshetnyak, 2012). Changes in the content or composition of bile acids (BAs) lead to the precipitation of cholesterol in bile, contributing to the formation of gallstones (Yu et al., 2021).

The gut microbiota can modulate the activation of the farnesoid X receptor (FXR) pathway signaling by altering the BA pool (Ye et al., 2023). FXR is a bile acid sensor that plays a crucial role in regulating cholesterol and bile acid homeostasis in the liver and gut (Forman et al., 1995; Mi et al., 2003; Wittenburg et al., 2003). Studies have shown that FXR suppresses gene transcription that encodes cholesterol-7-hydroxylase, which is a rate-limiting

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enzyme for bile acid synthesis (Doggrell, 2005). Experimental research demonstrated that FXR agonists have a protective effect against the progression of CGD, suggesting their potential use as therapeutic agents (Moschetta et al., 2004).

Currently, laparoscopic cholecystectomy is the primary treatment for CGD, which is one of the most commonly performed surgical procedures worldwide. However, cholecystectomy is invasive and can result in serious surgical complications. Moreover, not all CGD patients are suitable candidates for this procedure (Barahona Ponce et al., 2021; Lammert & Miquel, 2008; Shabanzadeh et al., 2016). Fortunately, noninvasive medical treatment options are available, such as oral litholysis with bile acids (BAs) (Portincasa et al., 2009). Ursodeoxycholic acid (UDCA) is the only FDA-approved BA for the treatment of CGD and has proven efficacy in gallstone dissolution (Guarino et al., 2013; Machado et al., 2019). Although the exact mechanism by which UDCA exerts its action is poorly understood, available data suggests that UDCA acts as a partial agonist via an FXR-mediated mechanism (Campana et al., 2005). Given the limited availability of approved drugs for nonsurgical CGD treatment and the high incidence of the disease, there is a significant market potential for better drugs (Claudel et al., 2005).

Computer-aided drug discovery (CADD) encompasses a wide range of theoretical and computational approaches that are used to identify and develop potential drug leads (Prieto-Martínez et al., 2019). CADD approaches have significantly expedited the drug discovery and development process, while also making it more cost-effective (Prasad et al., 2022). CADD approaches are divided into two categories, based on the availability of a 3D structure for the protein target: structure-based drug design (SBDD) and ligand-based drug design (LBDD) (Arya & Coumarb, 2021). SBDD relies on the presence of a 3D structure of the protein target, and the methods within this category aim to find and optimize small molecules that bind to the surface of this 3D structure (Klebe & Böhm, 1997). This study focuses on the discovery of new FXR agonists for the treatment of CGD using SBDD methods. The SBDD methods employed in this study include E-pharmacophore modeling, virtual screening, molecular docking, MM-GBSA calculations, and molecular dynamics (MD). These methods were used to evaluate compounds from the Zinc natural products library. Furthermore, the identified compounds were assessed for pharmacokinetics and toxicity.

Methods

All computational simulations, including molecular docking, MM-GBSA calculations, ADME analysis, and MD, were performed using Maestro v 12.8 from Schrödinger.

Protein preparation

The 3D structure of FXR and its co-crystalized ligand complex (PDB ID: 6HL1) was downloaded from the Protein Data Bank and prepared for docking using the protein preparation wizard panel of Maestro (Madhavi Sastry et al., 2013). The protein was preprocessed at pH 7.0/−2.0 by assigning bond orders, adding hydrogen atoms, assembling zero-order metal bonds, and converting selenomethionines to methionines. Prime was used to complete missing side chains and loops, and Epik was used to remove water except for 5 hetero groups to produce heat states. The protein was further refined by assigning H-bonds with structural optimization and sampling water orientations using PROPKA pH 7.0. The protein was minimized using the OPLS3 force field.

Ligand preparation

The Zinc database is an online resource that provides information on the properties and activities of a wide variety of natural compounds. In this study, we retrieved the structures of 270,540 compounds from the Zinc database. The ligand structures were then minimized using Schrodinger's Macromodel.

E-pharmacophore modeling and screening

The e-pharmacophore hypothesis was constructed using the FXR receptor-ligand complex with the "Develop a Pharmacophore from Receptor-ligand complex" option in Maestro's Phase. Energy functions were then applied to map the compounds, increasing the likelihood of optimal matching between the developed compounds and the protein receptor. The prepared library of natural compounds was screened using the pharmacophore-based approach, searching for matched compounds with pharmacophoric features (Salam et al., 2009).

Receptor grid generation, molecular docking, and free binding energy calculation (MM-GBSA) calculations

The active site of FXR was located and defined using the default options in Schrodinger's grid generation tool, which focused on the centroid of the preexisting bound ligand (Friesner et al., 2006). Docking was carried out using Schrodinger's Glide program with flexible ligand docking parameters, while the other parameters were kept at their default values. A hierarchical sequence of filters was used for the docking process. A library from Zinc database, which contains a vast number of natural compounds, was used for screening against the obtained pharmacophore. The obtained compounds were docked into the ligand binding pocket of FXR using the extra precision (XP) docking mode to minimize the occurrence of false-positive results. Finally, the top four compounds with the highest docking scores were selected for further MM-GBSA calculations.

To determine their binding affinity and the stability of the binding interactions, the top complexes generated from the XP mode were subjected to molecular mechanics with generalised Born and surface area solvation (MM-GBSA) calculations using the Prime module (Jacobson et al., 2002; Lyne et al., 2006). The OPLS3e force field, VSGB as the solvation model, and water as the solvent were used for the refinement. The results were compared with those of the co-crystallized ligand to evaluate the accuracy of the predicted binding affinities.

In silico ADMET prediction

In order to be considered a promising therapeutic candidate, any small molecule must possess favorable ADMET (i.e., Absorption, Distribution, Metabolism, Elimination, and Toxicity) properties. The physicochemical properties of the top-ranked ligands were evaluated using the QikProp module of Schrödinger. QikProp provides a range of parameters for comparing the properties of a given molecule to those of 95% of known drugs, including octanol/water partition coefficient, solubility coefficient, blood-brain barrier (BBB) permeability, HERG cardiac toxicity, Caco and MDCK cell permeability, and percent of human oral absorption (Das et al., 2021). It also assesses the suitability of compounds based on Lipinski's rule of five, which is essential for ensuring a drug-like profile during rational drug design.

Molecular dynamics simulations

MD simulations were used to examine the stability of protein-ligand complexes. The top-scoring compounds from XP docking were investigated with MD simulations for 100 ns using Academic Desmond v6.5. The complexes were neutralized by adding Na^+ and Cl^- ions and immersed in an orthorhombic box of TIP3P solvent molecules. The simulations were performed at 300K and 1 bar pressure, and 1000 frames were collected for each system. After structure stabilization, the final models were obtained by averaging snapshots from the trajectory generated by MD simulations. The RMSD was computed to determine the average change in the location of selected atoms in a compound relative to the reference trajectory frame. The simulations were performed to ensure that the compounds met the criteria for protein-ligand equilibrium.

Results and discussion

The findings of the study have been illustrated in Fig. 15.1, which depicts the workflow followed for the investigation.

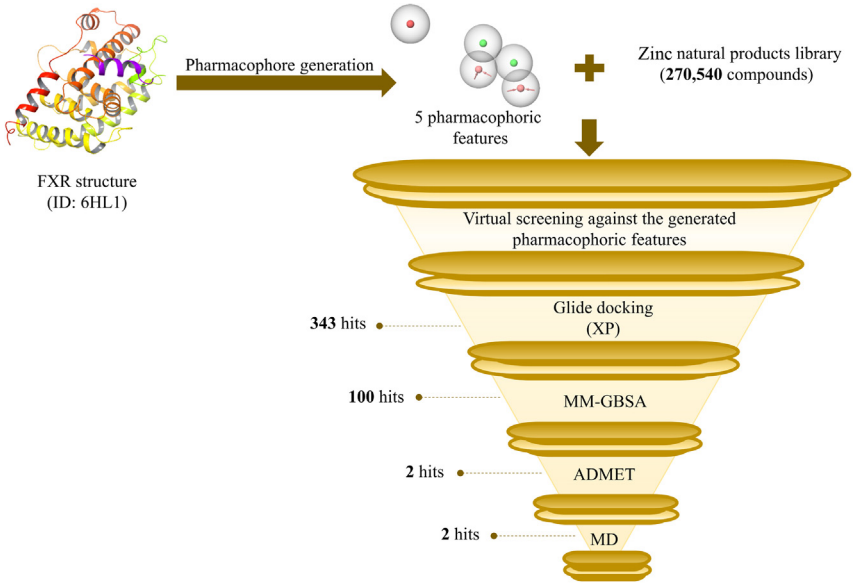

Figure 15.1 ***The overall work summary.*** This figure depicts all the computational methods used in this study.

E-pharmacophore generation and virtual screening

To start, an e-pharmacophore model was created based on the FXR and its co-crystalized ligand chenodeoxycholic acid (CDCA) (PDB ID: 6HL1). Notably, CDCA, a hydrophilic bile acid, and the most potent FXR agonist, was the co-crystallized ligand, although it is not currently used in clinical settings due to serious side effects (Jungsuwadee & Vore, 2010). The e-pharmacophore generation process resulted in five pharmacophoric features, including one negative ionic, two acceptors, and two hydrophobic groups (Fig. 15.2). These features were then used to screen the ZINC natural products library, which consisted of 270,540 compounds. From this screening, 343 compounds matched all five features and were subsequently subjected to molecular docking.

Molecular docking and MM-GBSA calculations

Molecular docking plays a crucial role in computer-aided drug discovery by predicting the binding affinity and the interactions between a receptor and a ligand. It can identify the residues of the receptor that interact with the ligand and the types of interactions formed (Obubeid et al., 2022). In this study, the FXR active site was used to perform molecular docking in extra precision (XP) mode on the 343 compounds that passed the pharmacophore-based virtual screening. The top 100 compounds were identified with docking scores ranging from −16.294 to −11.104 Kcal/mol, while the co-crystallized

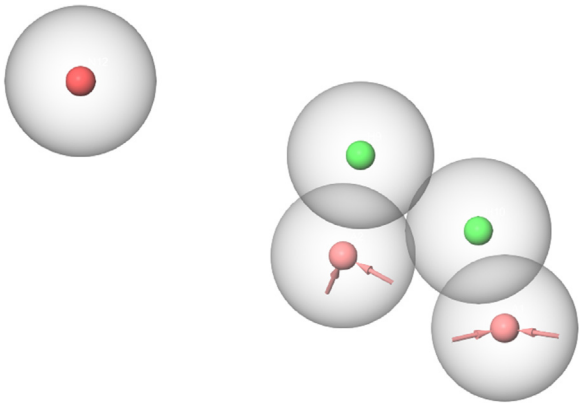

Figure 15.2 ***Pharmacophore hypothesis developed with FXR (PDB ID: 6HE1).*** The hypothesis was generated using the "Generate hypothesis from receptor-ligand complex" option of the Phase module. Pink sphere with arrows, hydrogen-bond acceptor; red sphere, negative ionic; green sphere, hydrophobic.

ligand CDCA achieved a docking score of −15.901 Kcal/mol. Additionally, UDCA was docked against the FXR active site and attained a docking score of −14.063 Kcal/mol.

The top 100 compounds, as well as CDCA and UDCA, underwent free-binding energy calculations using the MM-GBSA method. This method considers the impact of the solvent on the binding of the ligand to the receptor (Alzain, 2022). Four compounds, namely ZINC000253500120, ZINC000013543173, ZINC000085541241, and ZINC000035363836, were selected for further analysis. These compounds achieved MM-GBSA dG bind energies of −52.01, −84.50, −70.97, and −59.42 Kcal/mol, and docking scores of −15.272, −14.978, −14.517, and −14.462 Kcal/mol, respectively (Table 15.1). CDCA and UDCA, the two FXR natural agonists, obtained MM-GBSA dG bind energies of −90.68 and −84.18 Kcal/mol, respectively. The selection of the four compounds was based on both their MM-GBSA dG bind energies and their previously calculated docking scores. It is evident that none of the four compounds obtained a superior docking score to CDCA (−15.90), although all of them outperformed UDCA (−14.063). Similarly, none of the compounds exhibited a higher MM-GBSA dG bind energy than CDCA (−90.68). However, ZINC000013543173 was the only compound with a superior MM-GBSA dG bind energy than UDCA (−84.50 and −84.18, respectively), while the other compounds still produced satisfactory results.

The top two compounds from Table 15.1, ZINC000253500120 and ZINC000013543173, were selected for further investigation of their interaction patterns in comparison with CDCA and UDCA, in order to gain insight into their potential as FXR modulators (Fig. 15.3). Interestingly, all four compounds formed similar interactions with the receptor, including a

Table 15.1 Docking scores and MM-GBSA dG bind energies of the four selected best ligands poses in addition to CDCA and UDCA.

Compound	Docking score (Kcal/mol)	MM-GBSA dG bind (Kcal/mol)
ZINC000253500120	−15.272	−52.01
ZINC000013543173	−14.978	−84.50
ZINC000085541241	−14.517	−70.97
ZINC000035363836	−14.462	−59.42
CDCA	−15.901	−90.68
UDCA	−14.063	−84.18

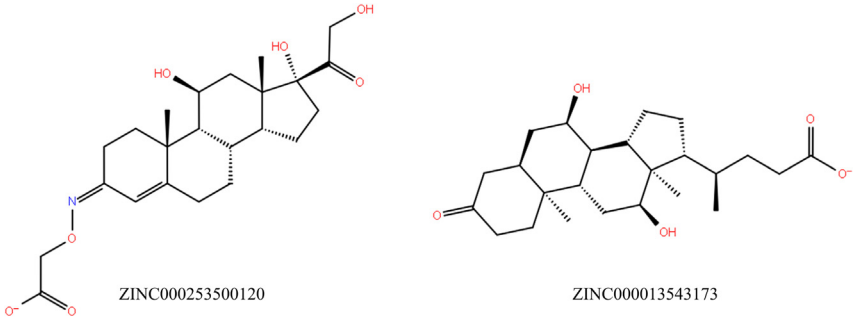

Figure 15.3 ***2D structures of the top two compounds.*** ZINC000253500120 and ZINC000013543173 structures.

salt bridge with ARG331, H-bonds with HIS447, and hydrophobic contacts with multiple residues including ALA291, MET290, LEU287, TRP454, TYR361, ILE362, MET365, MET450, ILE357, TRP469, ILE352, MET328, PHE329, LEU348, TYR369, ILE335, PHE336, and MET265. Additionally, all compounds showed polar interactions with HIS447, SER332, and HIE294, as well as a charged positive interaction with ARG331. However, there were some differences in interactions observed between the compounds. Specifically, ZINC000253500120, ZINC000013543173, and CDCA formed H-bonds with SER332, while ZINC000013543173 and CDCA formed H-bonds with TYR369. Furthermore, ZINC000013543173, CDCA, and UDCA made hydrophobic contacts with PHE366 and LEU451 (Table 15.2 and Fig. 15.4). The comparison of ZINC000253500120 and ZINC000013543173 with CDCA and UDCA reveals that they have a high potential to act as FXR modulators since they interact with the receptor in a similar way. The presence of H-bond and hydrophobic interactions plays a crucial role in predicting binding affinity and FXR recognition (Chen et al., 2016; Klebe & Böhm, 1997; Wei et al., 2021). The compounds and references show a similar pattern of interactions with only a few exceptions. Therefore, it can be hypothesized that the hydrophobic contacts with PHE366 and LEU451 contributed to the increase in MM-GBSA dG bind energy. ZINC000013543173 and the two references that have comparable MM-GBSA dG bind energy exhibited these interactions, whereas ZINC000253500120, which has a more positive dG bind energy, lacks these interactions. CDCA has a higher docking score due to possessing four H-bonds interactions, which is one more than our two compounds that only made three H-bonds. Likewise, ZINC000253500120 and ZINC000013543173 have a higher docking score

Table 15.2 Ligand-residue interactions of ZINC000253500120, ZINC000013543173, CDCA, and UDCA with FXR.

Compound	Salt bridge	H-bond	Hydrophobic	Other interactions
ZINC000253500120	ARG331	HIS447, TYR361, SER332	ALA291, MET290, LEU287, TRP454, TYR361, ILE362, MET365, MET450, ILE357, TRP469, ILE352, MET328, PHE329, LEU348, TYR369, ILE335, PHE336, MET265	**Polar:** HIS447, SER332, HIE294 **Charged positive:** ARG331
ZINC000013543173	ARG331	HIS447, TYR369, SER332	ALA291, MET290, LEU287, TRP454, TYR361, ILE362, MET365, MET450, ILE357, TRP469, ILE352, MET328, PHE329, LEU348, TYR369, ILE335, PHE336, MET265, PHE366, LEU451	**Polar:** HIS447, SER332, HIE294 **Charged positive:** ARG331
CDCA	ARG331	HIS447, TYR361, SER332, TYR369	ALA291, MET290, LEU287, TRP454, TYR361, ILE362, MET365, MET450, ILE357, TRP469, ILE352, MET328, PHE329, LEU348, TYR369, ILE335, PHE336, MET265, PHE366, LEU451	**Polar:** HIS447, SER332, HIE294 **Charged positive:** ARG331

(*Continued*)

Table 15.2 (Continued)

Compound	Salt bridge	H-bond	Hydrophobic	Other interactions
UDCA	ARG331	HIS447, TYR361	ALA291, MET290, LEU287, TRP454, TYR361, ILE362, MET365, MET450, ILE357, TRP469, ILE352, MET328, PHE329, LEU348, TYR369, ILE335, PHE336, MET265, PHE366, LEU451	**Polar:** HIS447, SER332, HIE294 **Charged positive:** ARG331

than UDCA, which has only two H-bonds, because they made three H-bonds. However, the difference is not significant.

Furthermore, a literature review was conducted to investigate if the observed interactions with ZINC000253500120 and ZINC000013543173 were previously reported. The analysis revealed that the most potent compounds in several studies share similar interactions with our two hits. For instance, Sepe et al., identified an isoxazole derivative, which was found to form a salt bridge interaction with ARG331 and H-bond with HIS447 (Sepe et al., 2019). Notably, the interaction with ARG331 was shown to stabilize the binding between bile acid derivatives and FXR. (−)-fargensone A (FA), a natural product discovered by Guo et al., made H-bond with HIS447 and hydrophobic contacts with LEU287, ILE352, and TRP454, which were mentioned to stabilize FA binding (Guo et al., 2022). Wei group isolated and evaluated another natural product, dehydrodicoferyl product (DA), which was demonstrated to form significant hydrophobic interactions with ILE352, LEU287, and LEU348 (Wei et al., 2021). Additionally, a synthetic UDCA derivative called 6EUDCA was synthesized and tested for its FXR agonistic activity via the Yu group. 6EUDCA exhibited H-bonds with HIS447 and SER332 and strong hydrophobic contact with TRP454 (Yu et al., 2016). This analysis suggests that compounds ZINC000253500120 and ZINC000013543173 have a high potential for being promising FXR modulators as they share many common interactions with potent FXR modulators. Furthermore, two of the potent FXR modulators that were reviewed are natural compounds, which enhances the convenience of our library selection.

ADMET prediction

The poor pharmacokinetics profile is one of the major reasons why many hits fail to reach the final stages of drug development. Fortunately, the ADMET properties of compounds can be predicted using computational tools such as QikProp, which can help in lowering the failure probability in future drug development stages (Abd Emoniem et al., 2023; AbdElmoniem et al., 2023). In this study, we predicted the pharmacokinetic properties of ZINC000253500120 and ZINC000013543173, along with the references CDCA and UDCA (Table 15.3). QPlogPo/w parameter was used to represent the lipophilicity of the compounds, and their values (1.396, 2.909, 3.826, and 3.802, respectively) showed acceptable lipophilicity. The aqueous solubility was represented by the

QPlogS parameter, and the values achieved by the compounds (−3.635, −4.518, −5.102, and −5.105, respectively) demonstrate good solubility. QPPCaco and QPlogBB denote the likelihood of the compounds to cross the gut and blood-brain barriers, respectively. UDCA has the highest gastrointestinal absorption with a QPPCaco of 45.922, followed by CDCA with a value of 39.972, then ZINC000013543173 (23.880), and finally,

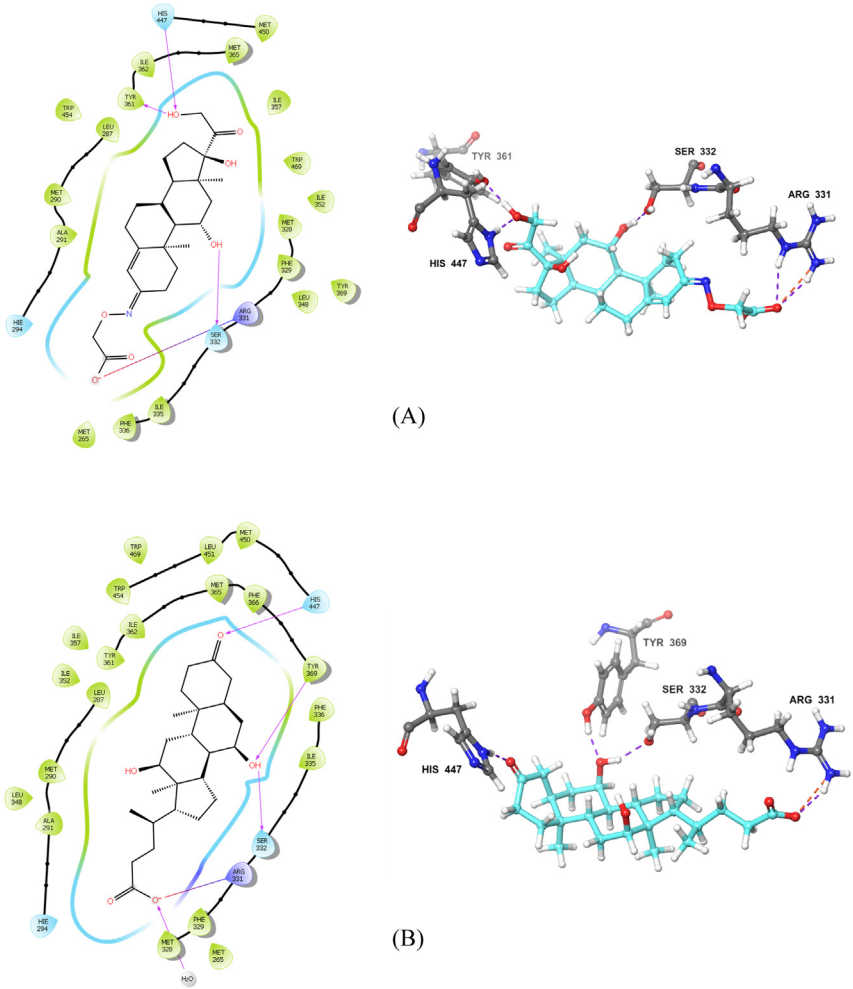

Figure 15.4 ***2D and 3D interactions of the top two compounds and the two references (CDCA and UDCA) with FXR (PDB ID: 6HE1) using the Glide software.*** (A) ZINC000253500120 (cyan). (B) ZINC000013543173 (cyan). (C) CDCA (green). (D) UDCA (green).

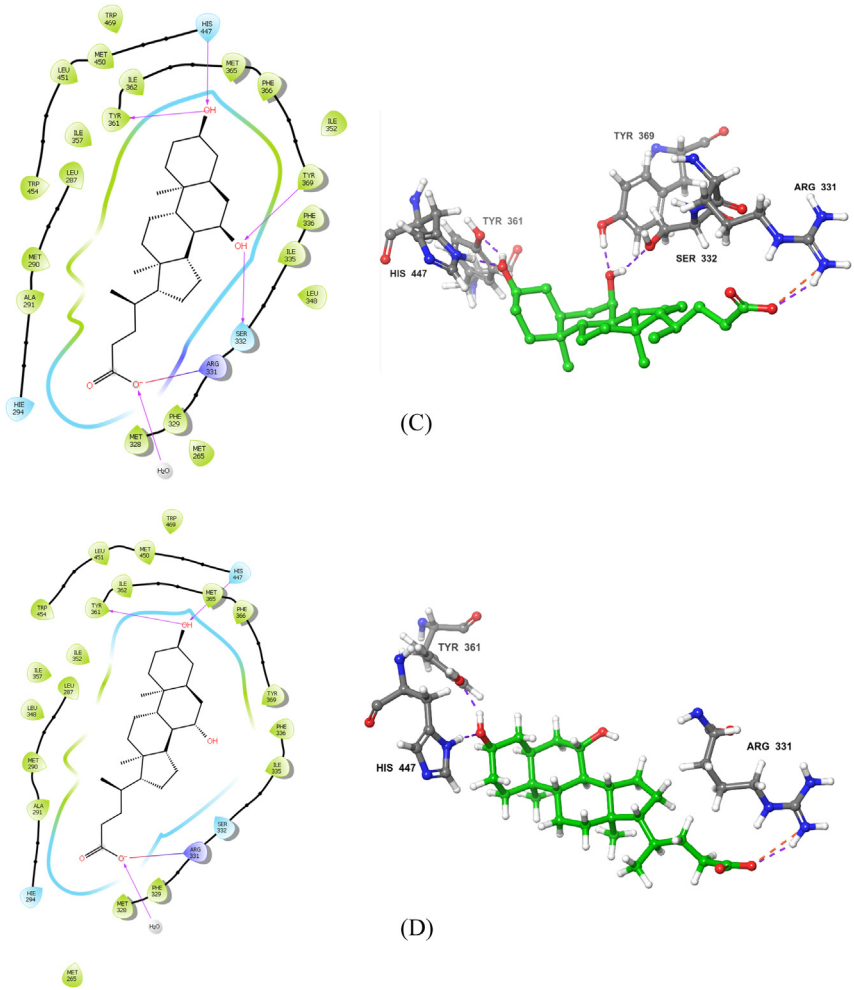

Figure 15.4 ***(Continued)***

ZINC000253500120 (9.860). The low QPlogBB values achieved by all compounds and references (≤ −2.413) predict limited central nervous system (CNS) penetration. None of the compounds or references showed cardiotoxicity, and none of them deviated from the rule of five. The human oral absorption percent for ZINC000253500120, ZINC000013543173, CDCA, and UDCA was found to be 52.909%, 68.642%, 79.093%, and 77.876%, respectively, which is in agreement with their respective values of 2, 2, 3, and 3 (1, 2, or 3 indicated low,

Table 15.3 ADME analysis of ZINC000253500120, ZINC000013543173, CDCA, and UDCA using the QikProp tool.

Compound	QPlogPo/w	QPlogS	QPlogHERG	QPPCaco	QPlogBB	Human oral absorption	%Human oral absorption	Rule of five
ZINC000253500120	1.396	−3.635	−2.521	9.860	−2.413	2	52.909	0
ZINC000013543173	2.909	−4.518	−1.968	23.880	−1.714	2	68.642	0
CDCA	3.826	−5.102	−2.079	45.922	−1.471	3	79.093	0
UDCA	3.802	−5.105	−2.060	39.972	−1.524	3	77.876	0

medium, and high absorption). The ADMET analysis suggests that ZINC000253500120 and ZINC000013543173 have good pharmacokinetic profiles that qualify them for further investigation.

Molecular dynamics

In the human body, proteins and molecules are in a continuous state of motion and dynamics. However, molecular docking assumes rigid structures for proteins and compounds, neglecting their flexibility and movement. To address this limitation, MD simulation can simulate the behavior of proteins and compounds in the body under realistic conditions, accounting for conformational changes and reducing false positives obtained from molecular docking (Aghajani et al., 2022; Jordaan et al., 2020; Kumar et al., 2019). In this study, ZINC000253500120, ZINC000013543173, and UDCA were subjected to MD simulation to assess their stability and binding contacts with FXR. Root Mean Square Deviation (RMSD), Root Mean Square Fluctuations (RMSF), and protein-ligand contacts were analyzed over a simulation duration of 100 ns.

The average RMSD of the protein Cα atoms in complex with ZINC000253500120, ZINC000013543173, and UDCA was found to be 3.399817 Å, indicating comparable stability of the protein in the presence of the three ligands. This value is within the acceptable range of 1−3 Å for globular proteins, suggesting that the protein maintained its stability during the simulation. Additionally, as depicted in Fig. 15.5, the protein did not exhibit significant fluctuations during the simulation, further confirming its stability. Among the three ligands, ZINC000013543173 was the most stable, demonstrating tight binding with FXR for most of the simulation time. In contrast, ZINC000253500120 and UDCA showed higher fluctuations during the simulation period.

The small RMSF value indicates low flexibility and high stability. All three complexes achieved the same average RMSF value of 1.455187 Å, which confirms their stability. As shown in Fig. 15.6, the protein displayed a similar pattern with the three ligands, indicating that they influenced the protein in the same manner. This suggests that ZINC000253500120 and ZINC000013543173 are FXR agonists similar to UDCA.

Fig. 15.7 shows the interaction fraction histograms of the ZINC000253500120, ZINC000013543173, and UDCA complexes with FXR. It is important to note that some residues had contact fractions exceeding 100%, which is due to these residues making multiple contacts

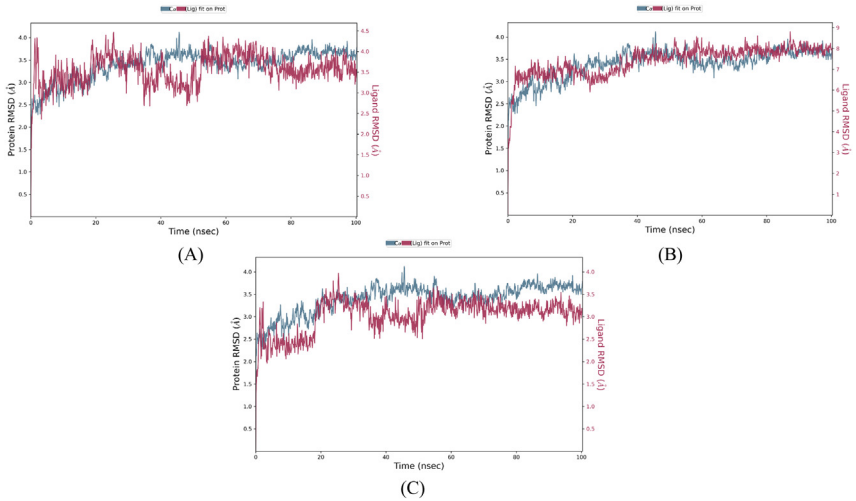

Figure 15.5 ***The protein-ligand RMSD plot of the top two compounds and UDCA complexed with FXR (PDB ID: 6HE1) during 100 ns molecular dynamics simulation using Desmond software.*** (A) ZINC000253500120. (B) ZINC000013543173. (C) UDCA.

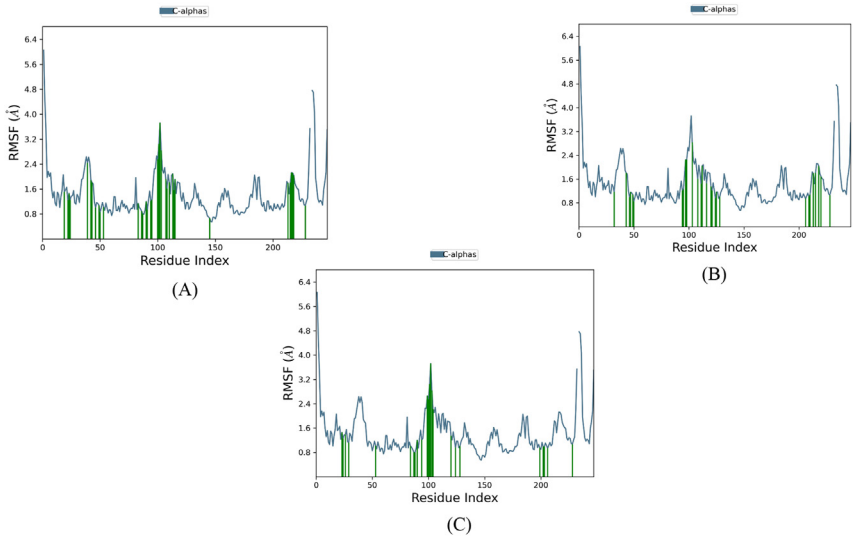

Figure 15.6 ***Plot of Protein RMSF showing the top two compounds and UDCA complexed with FXR (PDB ID: 6HE1) during 100 ns molecular dynamics simulation using Desmond software.*** (A) ZINC000253500120. (B) ZINC000013543173. (C) UDCA.

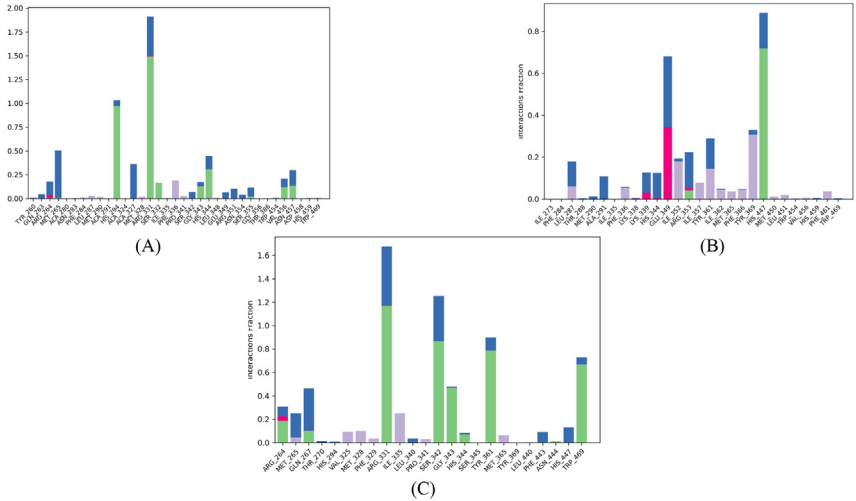

Figure 15.7 ***Protein-ligand contact histogram of the top two compounds and UDCA complexed with FXR (PDB ID: 6HE1) during 100 ns molecular dynamics simulation using Desmond software.*** (A) ZINC000253500120. (B) ZINC000013543173. (C) UDCA.

of the same interaction type with the ligand. The ZINC000253500120/FXR complex made contacts with MET265 (water bridge 50%), HIS294 (H-bond 90% and water bridge 10%), ALA327 (water bridge 35%), ARG331 (H-bond 145% and water bridge 40%), and HIS344 (H-bond 30% and water bridge 15%). The ZINC000013543173/FXR complex contacted with GLU349 (ionic 30% and water bridge 40%), TYR361 (hydrophobic 10% and water bridge 20%), TYR369 (hydrophobic 30% and water bridge 1%), and HIS447 (H-bond 70% and water bridge 20%). The UDCA/FXR complex exhibited contacts with GLN267 (H-bond 10% and water bridge 40%), ARG331 (H-bond 120% and water bridge 50%), SER342 (H-bond 85% and water bridge 35%), GLY343 (H-bond 45% and water bridge 1%), TYR361 (H-bond 80% and water bridge 10%), and TRP469 (H-bond 60% and water bridge 5%).

Conclusion

To discover new FXR modulators for the treatment of CGD, we used multiple computational approaches. First, a pharmacophore model was generated based on the receptor-ligand complex of the FXR crystal structure (ID: 6HL1), identifying five pharmacophoric features. The Zinc natural

products library (270,540 compounds) was screened against these features, yielding 343 compounds. Molecular docking and MM-GBSA calculations were then performed on these compounds, resulting in the selection of two compounds (ZINC000253500120 and ZINC000013543173) based on their docking score and MM-GBSA dG bind energy. A comparison of these compounds with reference compounds (CDCA and UDCA) and potent FXR agonists suggested that they are potential FXR modulators. ADMET prediction and MD simulation further confirmed that ZINC000253500120 and ZINC000013543173 have favorable pharmacokinetic profiles and good stability. Therefore, these compounds have the potential to be considered promising hits in FXR drug discovery and warrant further validation in wet laboratories.

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